🔑 **Answer Key — Teacher Copy · Lesson 6-6**

MSTAR PRACTICE WORKSHEET

Lesson 6-6: Identify Equivalent Algebraic Expressions

Grade 6 Mathematics · Standard 6.AT.7

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

One point each.

Question 1 SELECTED RESPONSE — CORRECT: A

Which expression is equivalent to 4x + 3x?

- **A** 7x
- **B** $7x^2$
- **C** 12x
- **D** 43x

4x + 3x = 7x. Combine the coefficients: 4 + 3 = 7.

Question 2 SELECTED RESPONSE — CORRECT: A

Which expression is equivalent to 2(m + 5)?

- **A** 2m + 10
- **B** 2m + 5
- **C** 7m
- **D** m + 10

2(m + 5) = 2m + 10 using the distributive property.

Question 3 SELECTED RESPONSE — CORRECT: B

Which property is shown: $3(x + 7) = 3x + 21$?

A Commutative

B Distributive

C Associative

D Identity

Multiplying 3 by each term inside the parentheses is the distributive property.

Question 4 SELECTED RESPONSE — CORRECT: A

Simplify: $5n + 2n$

A $7n$

B $7n^2$

C $10n$

D $52n$

$5n + 2n = 7n$. Add the coefficients: $5 + 2 = 7$. The variable stays as n.

Question 5 SELECTED RESPONSE — CORRECT: A

Producer A wrote $2(x + 3)$ for a beat's cost. Which expression is equivalent?

A $2x + 6$

B $2x + 3$

C $2x + 5$

D $x + 6$

Distribute the 2 to each term inside: $2 \cdot x + 2 \cdot 3 = 2x + 6$. Check $x = 5$: $2(5 + 3) = 16$ and $2(5) + 6 = 16$.

Question 6 SELECTED RESPONSE — CORRECT: C

A track stacks 3x and then one more x. Which expression is equivalent to 3x + x?

A $3x^2$

B 3x

C 4x

D 4

3x + x is 3 groups of x plus 1 more group of x = 4x. Add the coefficients: 3 + 1 = 4. Check x = 3: 3(3) + 3 = 12 and 4(3) = 12.

Part B — Test-Format Questions

Scoring notes and distractor rationales below each item.

Question 7 TWO-PART QUESTION

Part A — correct answer: A

Which expression is equivalent to 4(x + 3)?

- (A) 4x + 12
- (B) 4x + 3
- (C) 7x
- (D) x + 12

> Use the distributive property to multiply 4 by each term inside the parentheses: 4 · x = 4x and 4 · 3 = 12, so the expression is 4x + 12.

- **B:** The 4 reached x but not the 3. Distribute it to both terms inside.
- **C:** The 4 and 3 were added and x attached. They are not like terms; the 4 multiplies each of them.
- **D:** The 4 reached the 3 but not the x. Distribute it to both terms inside.

Part B — correct answer: A

Which reasoning shows why your answer to Part A is correct?

- (A) Multiply 4 by each term inside the parentheses: 4 · x + 4 · 3 = 4x + 12. Testing x = 2 gives 4(2 + 3) = 20 and 4(2) + 12 = 20, so the two expressions match.
- (B) Multiply only the first term inside the parentheses and copy the second term: 4x + 3.
- (C) Add 4 and 3 because both are numbers, then attach the variable: 7x.
- (D) Multiply only the second term inside the parentheses and copy the first term: x + 12.

> Equivalent expressions give the same value for every substitution. Distributing 4 to both terms produces 4x + 12, and a test value confirms the match.

Question 8 WRITTEN RESPONSE · 2 POINTS

Type II Reasoning: Testing Only One Value

> Priya claimed that $2x + x$ and $2x^2$ are equivalent expressions. Her only check was $x = 0$. She wrote: $2(0) + 0 = 0$ and $2(0)^2 = 0$. Because both results were 0, she said the expressions are equivalent.

Explain Priya's misconception and give the correct simplified form of $2x + x$.

Model answer: Substituting 0 makes many different expressions come out to 0, so one test value proves nothing. Test a second value: at $x = 2$, $2x + x = 2(2) + 2 = 6$ but $2x^2 = 2(2)^2 = 8$. The values differ, so the expressions are not equivalent. Since $2x$ and x are like terms, $2x + x$ simplifies to $3x$.

2 points	Student explains that one test value, especially 0, is not enough and shows a second value where the expressions differ, and gives $3x$ as the simplified form of $2x + x$.
1 point	Student gives $3x$ as the simplified form but does not explain why a single test value is not enough.
0 points	Student agrees with Priya that the expressions are equivalent or gives an unrelated response.

Part C — Show Your Thinking

Model answer and scoring guidance.

Question 9 WRITTEN RESPONSE · 2 POINTS

A student says $2x + 3$ and $5x$ are equivalent because both equal 5 when $x = 1$. Is the student correct? Explain why testing one value is not enough to prove equivalence, and find a value of x where they give different results.

Model answer: The student is not correct. Both expressions really do equal 5 when $x = 1$, but equivalence means equal for EVERY value of x, and one lucky match proves nothing. Testing $x = 2$ gives $2(2) + 3 = 7$ while $5(2) = 10$ — that single counterexample settles it.

Award 2 points for a complete explanation with correct mathematics, 1 point for a correct answer with thin reasoning, 0 for an unrelated response.

Answer Key. Score written responses with the rubric and guidance above; award partial credit per the 1-point rows.